

A better poker solver before paid compute

Mathematical audit, paired pilots, and research priorities

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The decision

The offline trainer is materially more efficient, and an exact-terminal sampling change improves both short-run policy diagnostics. The next purchase should still be a bounded scaling experiment, not a “blowout” run expected to certify Approximate GTO. No research result or local pilot establishes that every gate can be reached within the \$15 budget.

Keep the new numerical, memory, and card-evaluation optimizations. Advance terminal integration to the next accuracy-focused pilot. Keep the research-derived checkdown baseline as a lower-memory alternative. Neither has been deployed to the website.

What improved locally

The optimized ordinary trainer reproduces the 400-round seed-A root policy after the lookup and evaluator changes. Elapsed time fell from 47.82 to 35.96 seconds; peak resident memory fell from 2.876 to 2.190GB: about 25% less time and 24% less RAM. Numerical support corrections cause small changes on seed B, so this is not a claim that all old artifacts remain identical.

Paired configuration	Root deviation A / B, bb	States A / B, million
Original PCS, 400 rounds	0.650 / 0.675	6.60 / 6.39
Optimized PCS, 600 rounds	0.556 / 0.560	9.69 / 9.33
Terminal integration, 400	0.463 / 0.486	10.01 / 9.39
Terminal integration, 800	0.407 / 0.384	19.06 / 18.87
Checkdown baseline, 800	0.482 / 0.481	13.01 / 12.32

Terminal integration lowers the 400-round root estimates by about 28% on both seeds. The lower-memory baseline uses 32–35% fewer states at 800 rounds, but gives worse root estimates. These are root diagnostics, not measured full-game exploitability reductions.

All pilots: 20bb, default sizing, fixed DCFR, zero averaging delay, seeds 26001/26002, 2,000 held-out deals, 256 samples per root class, and 2,000 action-EV evaluation deals. Training covers all streets; the pilot exports contain preflop policy rows only. Measurements are from an Apple M5 with 16GiB RAM, not the proposed cloud CPU.

The gates are not interchangeable

The selected terminal-integration 800-round pair remains experimental. Existing thresholds were not loosened to make it pass.

Metric	Current result	Existing target
Full-game exploitability	Unmeasured for this policy	User preference ~0.50bb total
Root local deviation	0.407 / 0.384bb	At most 0.10bb
Max cross-seed action MAE	8.28%	At most 5%
Primary-action agreement	64.50%	At least 85%
Aggregate action delta	5.27%	At most 3%
Median / p95 hand TV	22.90 / 40.76%	At most 20 / 35%
Held-out unknown lookups	11.51 / 11.68%	At most 5% diagnostic
Forced-root unknown lookups	16.84 / 16.81%	At most 5% diagnostic
EV SE at most 0.02bb	23.41 / 21.53% of reach	At least 95%
Release lookup / storage audit	Not completed	99.99% / at most 20GB

Root trained-combo coverage is effectively 100%. Held-out untrained lookups pass their 2.5% diagnostic, at 1.87/1.84%. Neither result establishes full-policy serving coverage. Full quantized export size and post-quantization probability validity remain unmeasured for this candidate.

The repository's full-hand responder currently consumes neural policies. Its older serving-model scores cannot be attached to these tabular checkpoints. Likewise, the user's later approximate 0.50bb total full-game target must not be compared to the separate root diagnostic or silently divided by two.

A legal responder's expected winnings are a lower bound on best-response value. A low upper confidence limit on those winnings is not an exploitability certificate. This distinction is explicit in [Lišý and Bowling, Equilibrium Approximation Quality of Current No-Limit Poker Bots, 2016 / AAAI 2017](#).

Our root evaluator also chooses and reports its best action using the same samples. Maximizing noisy estimates introduces upward selection bias. Its nominal confidence bounds are therefore descriptive, not selection-corrected certification. The evaluator is unchanged between the sampling-mode pilots; terminal integration changes training.

Fix the arithmetic, then reduce unnecessary randomness

The audit found an actual numerical defect: an absolute chip tolerance was also used for probability/regret support. A vector such as $[1e-12, 3e-12, 0]$ became uniform instead of $[0.25, 0.75, 0]$. The corrected normalization is scale-invariant for positive finite weights and rescales safely when their sum overflows. Positive range support is retained; chip-accounting tolerances are unchanged.

Terminal evaluation now caches hand ranks per sampled board and obtains compatible mass from total and per-card marginals, including exact card removal and ties. This is aligned with [Johanson et al., Efficient Nash Equilibrium Approximation through Monte Carlo CFR, AAMAS 2012](#). Sharing immutable trajectory/action slices reduces allocation without sharing regrets. Bit-parallel card ranking is checked against every one of the 8,192 rank masks.

Terminal integration evaluates opponent terminal branches exactly, then samples one remaining continuation with its conditional probability. The price is predictable: more continuation recursion means more persistent information sets.

The research follow-up instead builds stateless legal check/call-to-showdown branch baselines and samples the original proposal, so terminal selections still stop. If F is the uncorrected branch contribution, B its baseline, and q the proposal, the returned vector is:

$$\text{sum}(B) + (\text{sampled } F - \text{selected } B) / \text{selected } q$$

The child range already includes $1/q$, so the implementation does not divide its returned value again. Exact terminal baselines have zero residual. Algebraic tests cover every exact combo, nonuniform proposals, inaccurate baselines, and zero action-sampling variance with exact baselines.

This application uses the residual-correction identity in [Davis, Schmid and Bowling, Low-Variance and Zero-Variance Baselines for Extensive-Form Games, ICML 2020](#). Our particular stateless range-vector design is an adaptation, not a published guarantee about this solver. Poor baselines can increase variance; the empirical gains are mixed.

At 800 rounds, checkdown's aggregate action delta is better than integration's (2.54% versus 5.27%), but primary agreement is worse (57.99% versus 64.50%). It is retained for research, not automatically selected. No full-policy variance-times-cost advantage was established.

What could still prevent convergence to the desired policy?

More iterations reduce learning error in the represented game. They do not necessarily remove losses caused by buckets that combine strategically different hands or an action grid that omits profitable responses. The actual error floor of this repository's abstraction is unknown, not proven insurmountable.

The distinction is empirically illustrated in [Johanson et al., Finding Optimal Abstract Strategies in Extensive-Form Games, AAAI 2012](#): better abstract-game convergence can accompany worse full-game exploitability. Refining the abstraction is not automatically monotonic either, as shown by [Waugh et al., Abstraction Pathologies in Extensive Games, AAMAS 2009](#).

The most useful next policy work is therefore targeted, not a new network architecture by default:

1. Connect the existing legal multi-street responder to the frozen tabular average policy. Keep search and evaluation deals separate, maintain compatible opponent ranges, and score actual terminal returns. Use shared evaluation deals to compare candidates.
2. If losses persist despite better coverage, use those public states for a CFR-BR-inspired turn/river pilot against exact-card responses. Preserve the upstream ranges and continuation constraints. A local approximate implementation does not inherit a global CFR-BR guarantee.
3. Split only buckets shown to conceal conflicting action values. Test the resulting actions against held-out responses; do not enlarge the entire table on faith.
4. If table growth remains binding, generate exact turn/river targets for a small range-conditioned continuation model, including deliberately perturbed ranges. Judge the resulting decisions, not just average RMSE.

[DeepStack, Moravčík et al., Science 2017](#) motivates range-conditioned continuation values and constrained resolving. [ReBeL, Brown et al., NeurIPS 2020](#) reports using 90 eight-GPU DGX-1 machines for full-game data generation. Their full recipes are not credible \$15 reproductions.

[DREAM, Steinberger et al., 2020](#) still incurs a regret-error term from imperfect advantage fitting. A neural rewrite does not eliminate approximation error. Similarly, [Brown and Sandholm, Reduced Space and Faster Convergence via Pruning, ICML 2017](#) requires substantial early storage; it does not license deleting negative sampled-DCFR regrets here.

A bounded cloud decision, not a convergence promise

The updated runbook proposes a 1,600-round terminal-integration pair with a 45-million-information-set cap per seed, fixed DCFR, and checkpoints every 800 rounds. The prior 25M cap is too close to the new method's observed growth. Start a fresh schema-5 lineage; old schema-4 checkpoints require their original binary.

At the diagnostic allowance of 700 bytes per set, two workers and 20% headroom project 75.6GB (70.4GiB). This is a planning bound for preflop-only exports, not measured peak memory for complete exports or large checkpoint reloads. The generic full-export allowance remains 2,300 bytes per set. Keep both independent seeds; never merge their regrets.

Continue beyond that stage only when the observed policy/coverage trend and memory slope justify it, with paid time remaining. Before a multi-hour continuation, obtain full-hand responder evidence for this exact policy. The small pilots cannot establish a 12-hour completion time on Hetzner, and additional vCPUs do not parallelize a single deterministic solver seed.

No server was provisioned and no provider price was assumed. Before paid work, verify the selected machine, hourly price, taxes and ancillary charges against the total budget. An independent server-deletion/watchdog mechanism still needs verification. A solver stopping or a machine shutting down is not proof that cloud billing stopped.

Verification and evidence limits

The Rust release suite covers arithmetic, exact terminal settlement, legal trees, sampling expectation, descriptor independence, card-rank equivalence, and deterministic checkpoint resume. All three sampling modes are pinned across resume; older schemas and mode changes are rejected. Cloud-runner tests cover command construction, fingerprints, summary attestation and lineage.

The local cloud-runner smoke test uses actual complete-postflop exports and lossless checkpoint files. It exercises process orchestration only: its tiny policies are not quality results. No browser, API, web-app policy, or UI was changed.

Timing comparisons are local observations. Some larger pilots overlapped compilation or incurred memory-compression/kernel overhead; user CPU and instruction counts are retained alongside elapsed time. The roughly comparable 400-round integration versus 600-round ordinary control is not a rigorously equal-wall experiment. Root pilot samples are reused for model selection, so a final independent hold-out is still required.

The repository audit includes 17 measured runs, seven paired comparisons, binary/artifact hashes, settings, and remaining gates. Research used original papers and author publication pages, with consequential claims independently checked. Further broad searching was stopped because the remaining gaps require policy experiments, full-hand response evaluation, or actual host measurements.

Bottom line: the code is better prepared for a controlled paid pilot. It is not yet evidence-backed to promise an Approximate GTO release from one large training run.